
Central Nervous System

Introduction

Neuroscience

- *Significance:*
 - Sensory detection (+ sensory receptors):
 - Internal environment (pH, Glucose, ABP, Temp...etc.,)
 - External environment (special senses...)
 - Information processing (+ Learning & memory)
 - Transmission of information via neural networks.
 - chemical + electrical...
 - Transformation of signal... (neural integration)
 - Perception of sensory information...

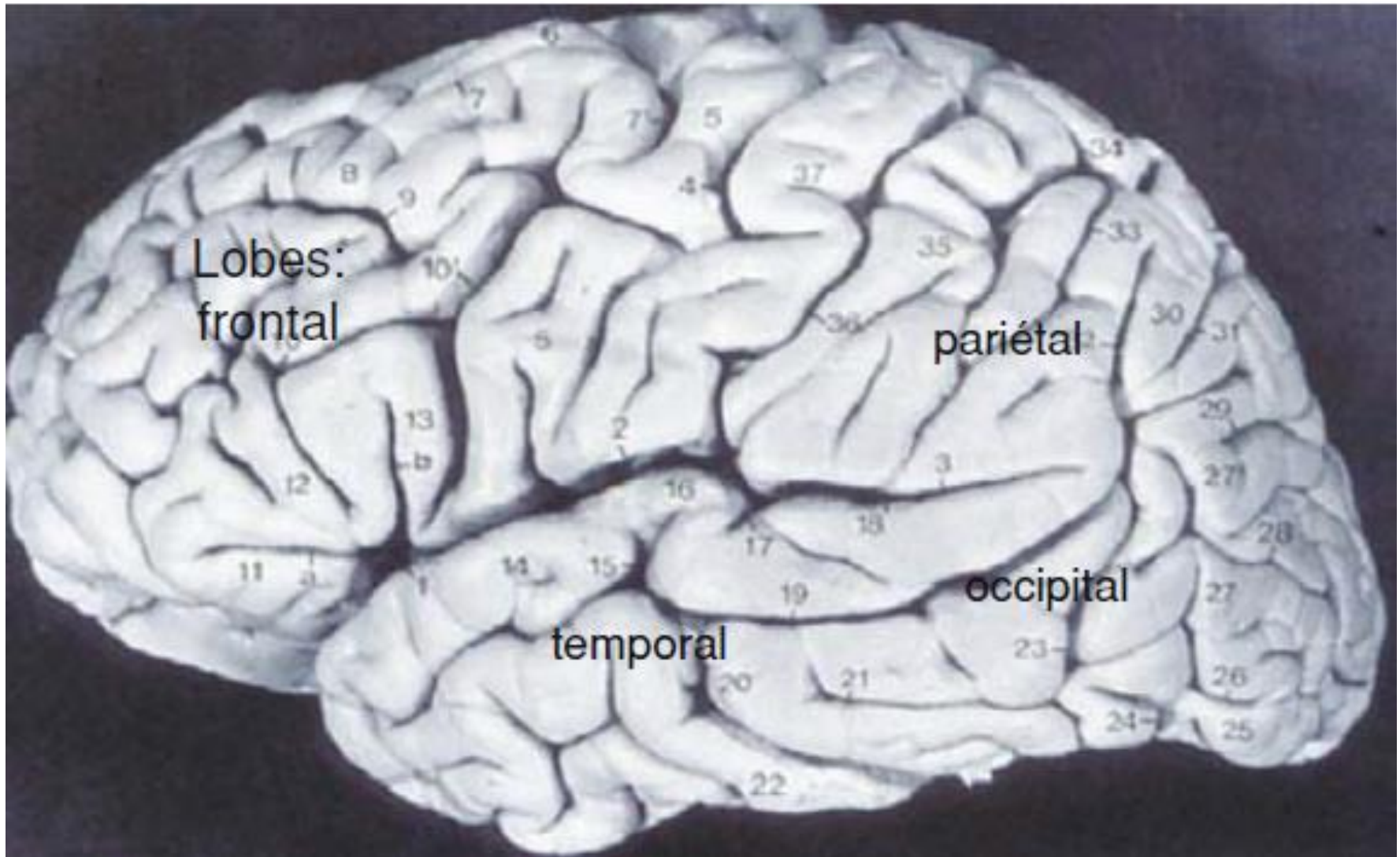
Neuroscience

- *Significance:*
 - Storage and retrieval of information (memory...)
 - Planning and implementation of motor commands
 - Thought processes and conscious awareness
 - Learning
 - Emotion and motivation...
- Expression of behavior...

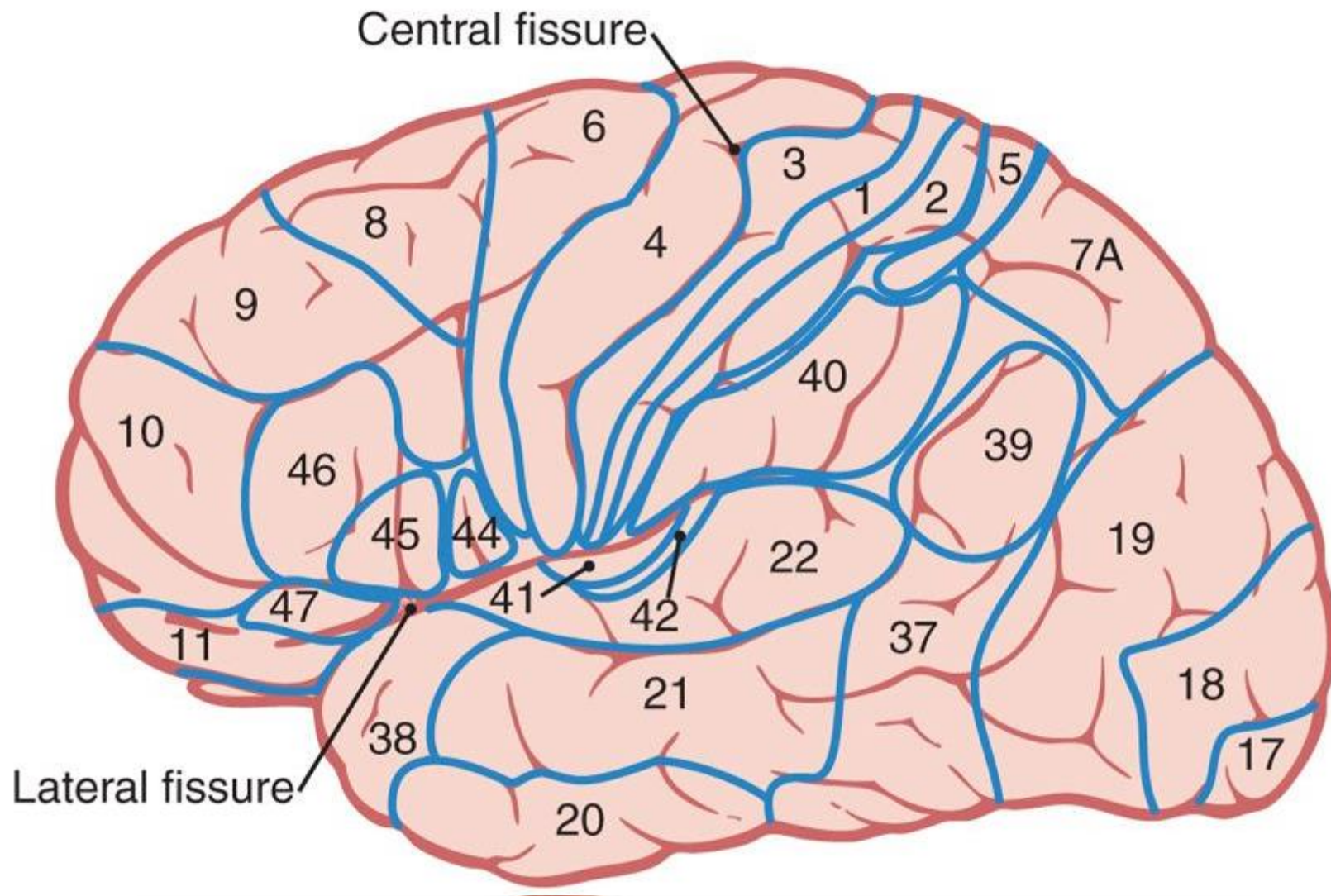
Neuroscience

- Human brain $\cong$ 1500gms
- Neuron $=$ 10^{12}
- 1 neuron forms $=$ 10^3 - 10^5 synaptic junctions
- Possible synapses $=$ $10^{15} - 10^{17}$

Cytoarchitectonic map



Cytoarchitectonic map



52 areas (+) anatomically and functionally distinct areas in human cerebral cortex

Nervous System: **CNS + PNS**

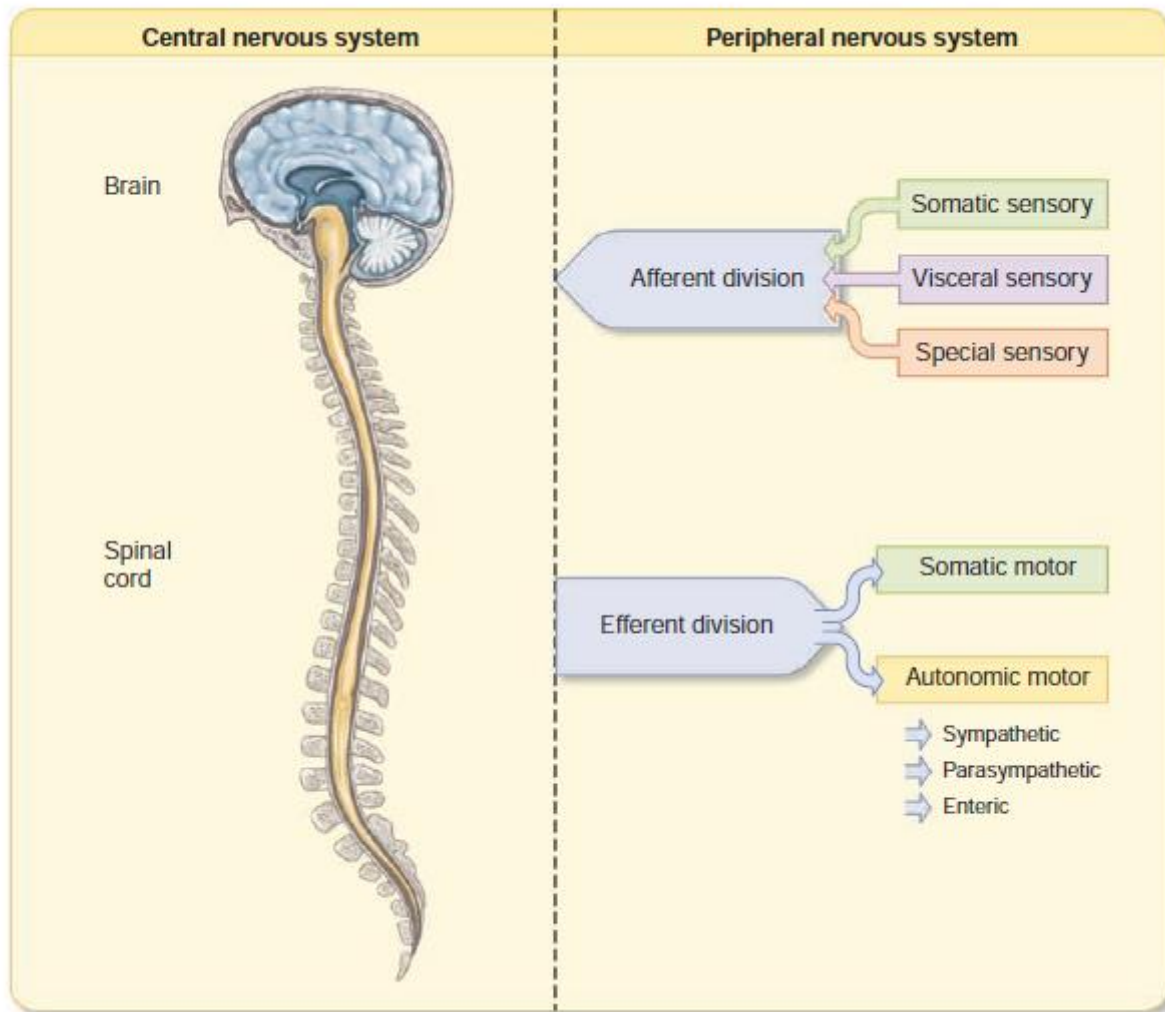
7 main parts: *Brain + SC*

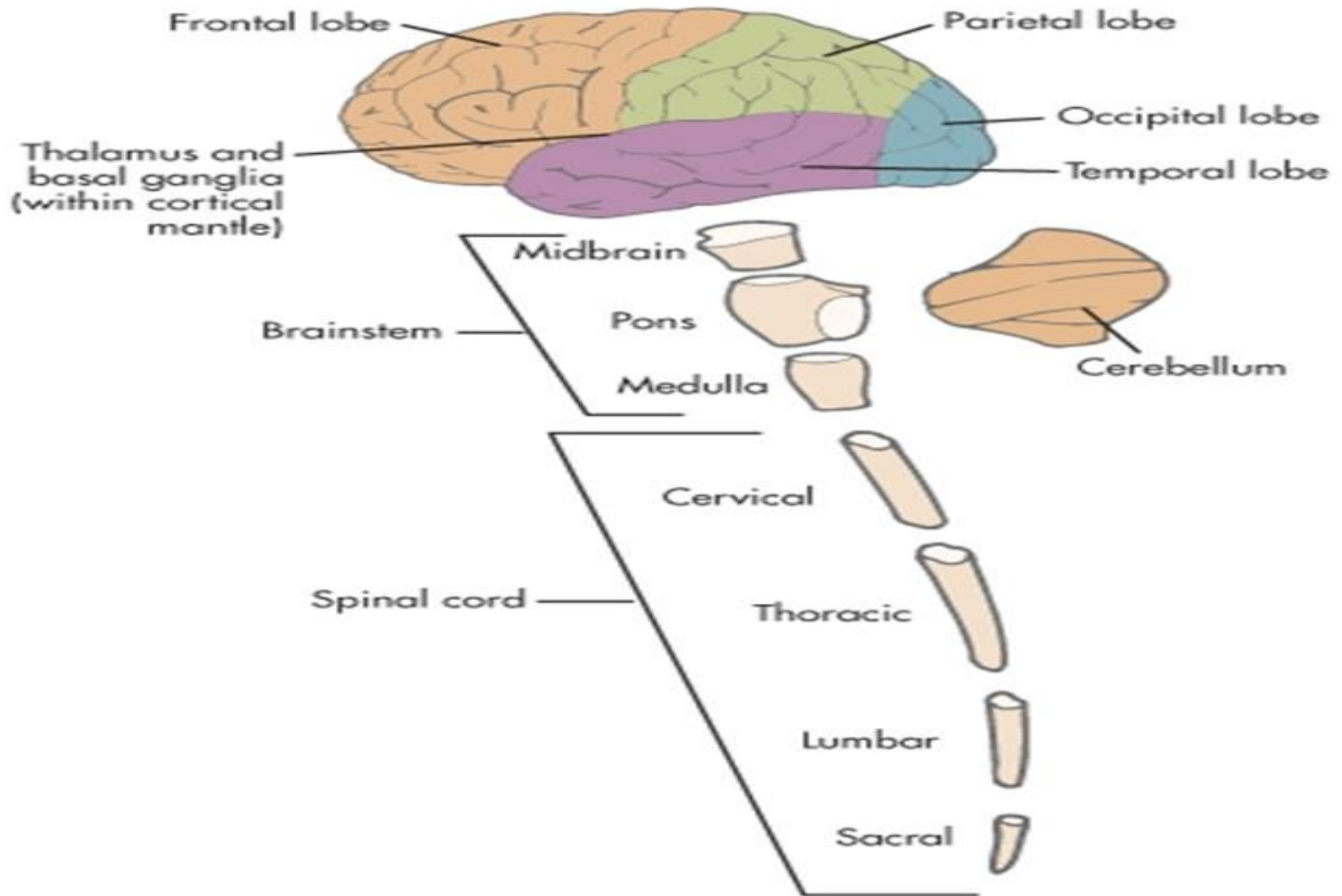
1. *Spinal cord*
2. *Medulla oblongata*
3. *Pons*
4. *Cerebellum*
5. *Midbrain*
6. *Diencephalon*
7. *Cerebrum*

Brain:

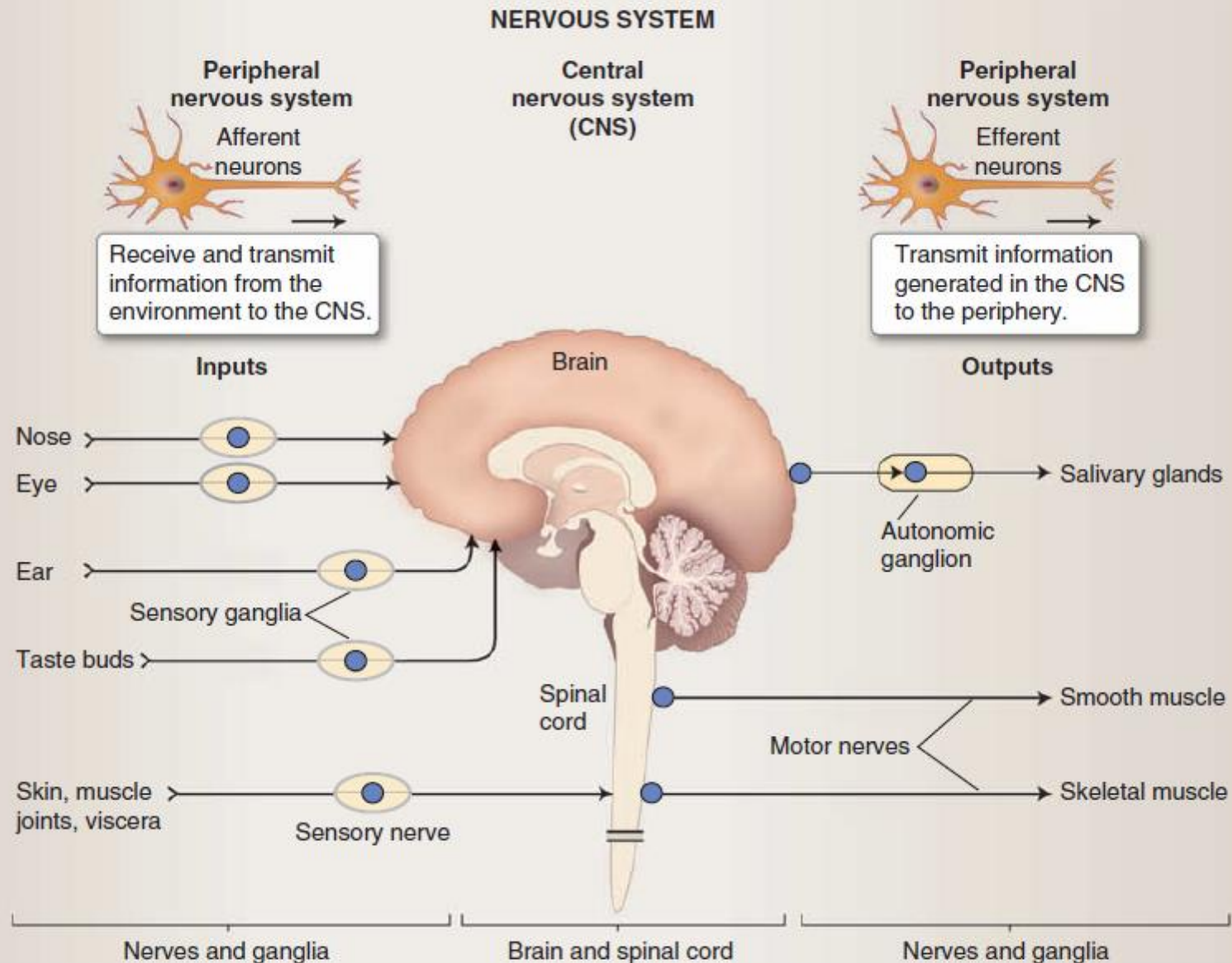
1. *Hindbrain : Medulla oblongata*
Pons
Cerebellum
2. *Midbrain*
3. *Forebrain (diencephalon + CC)*
 - *Medulla + Pons + Midbrain = brain stem.*

Components of the CNS

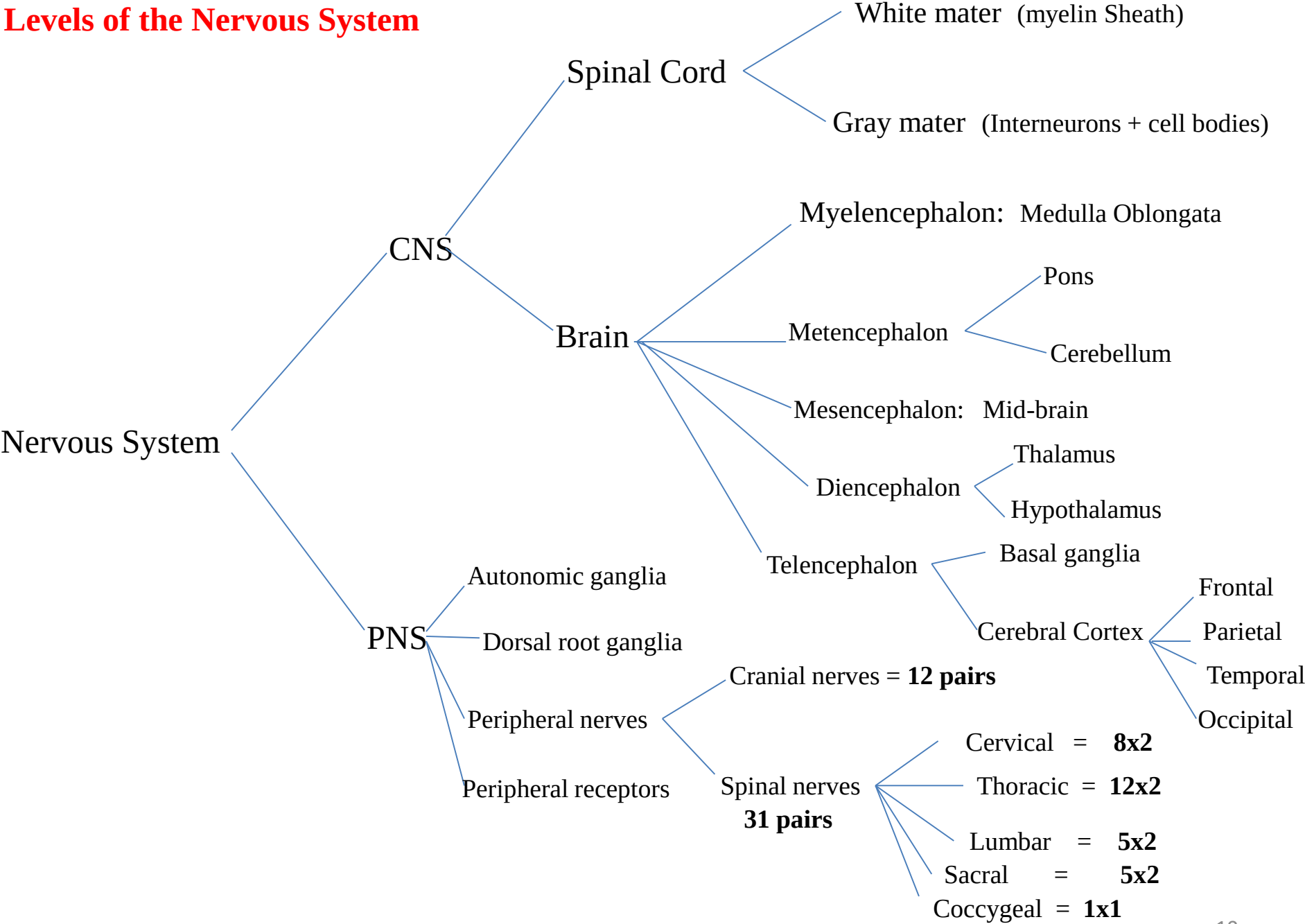


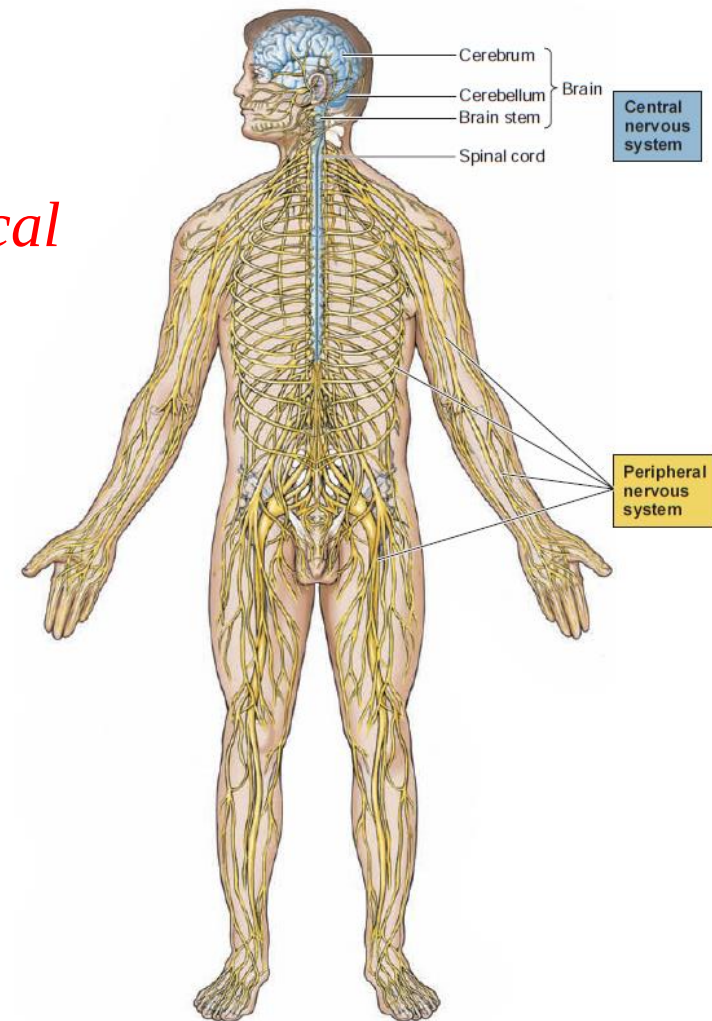
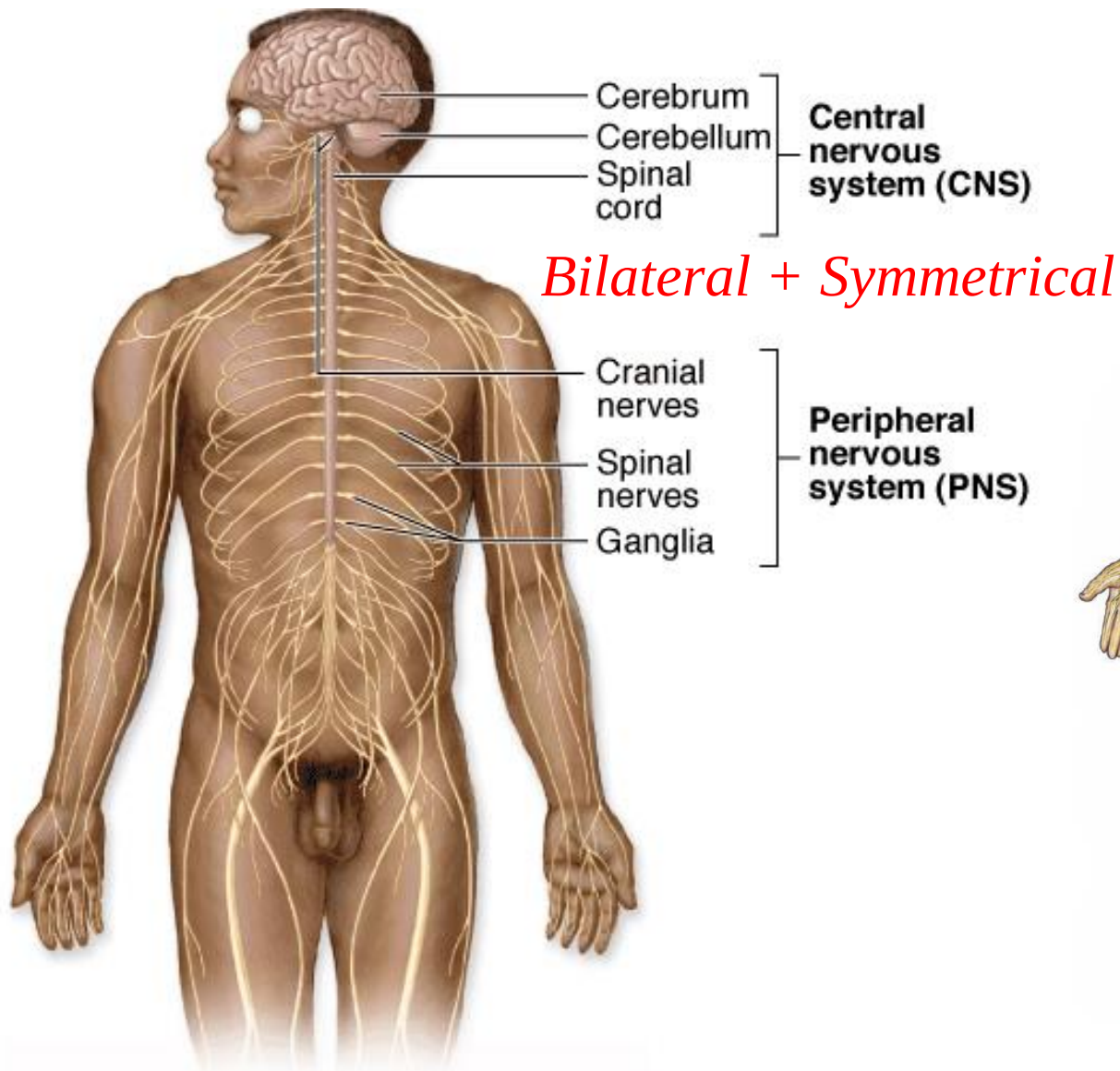


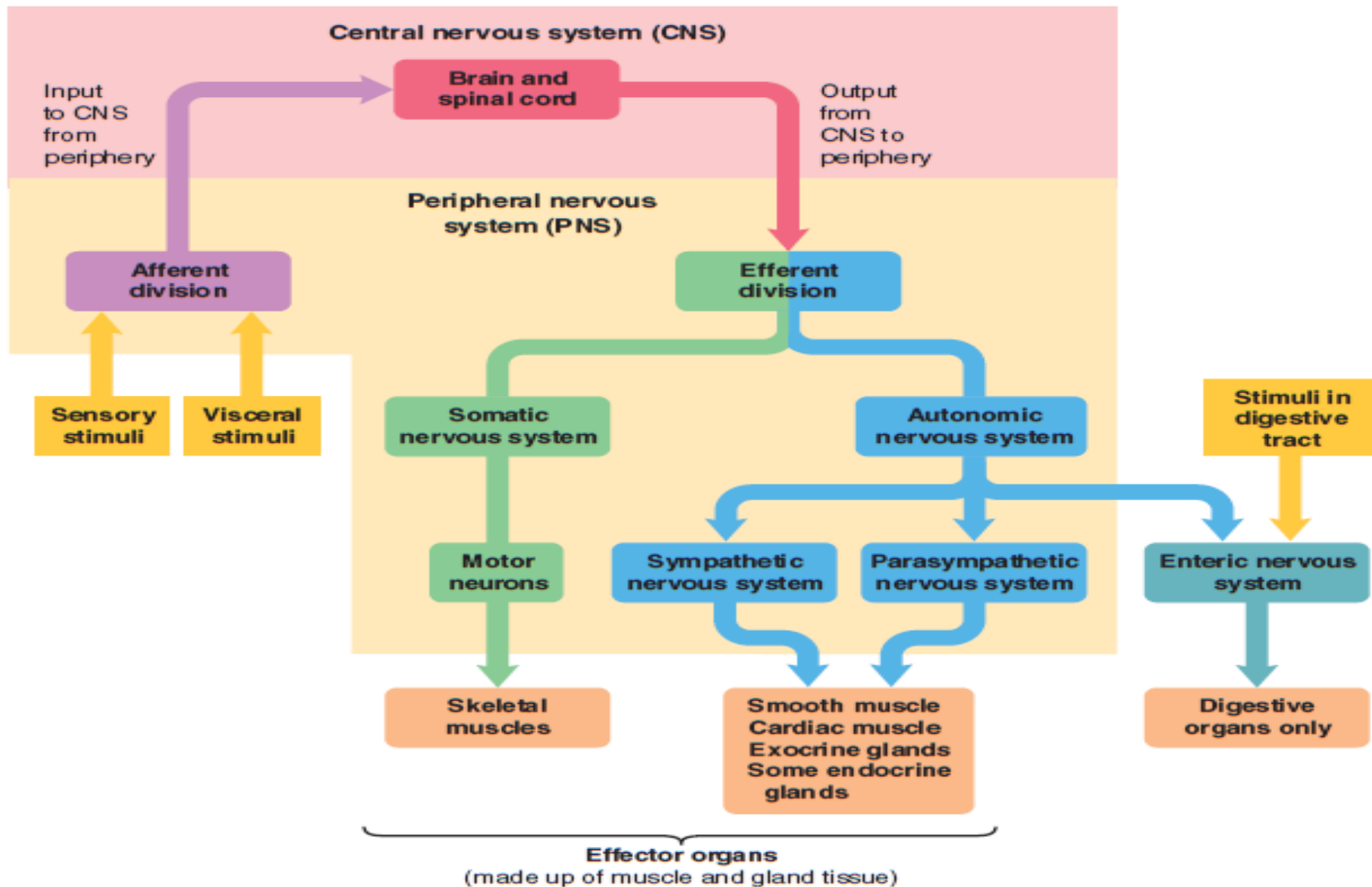
Components of Human Brain and Spinal Cord



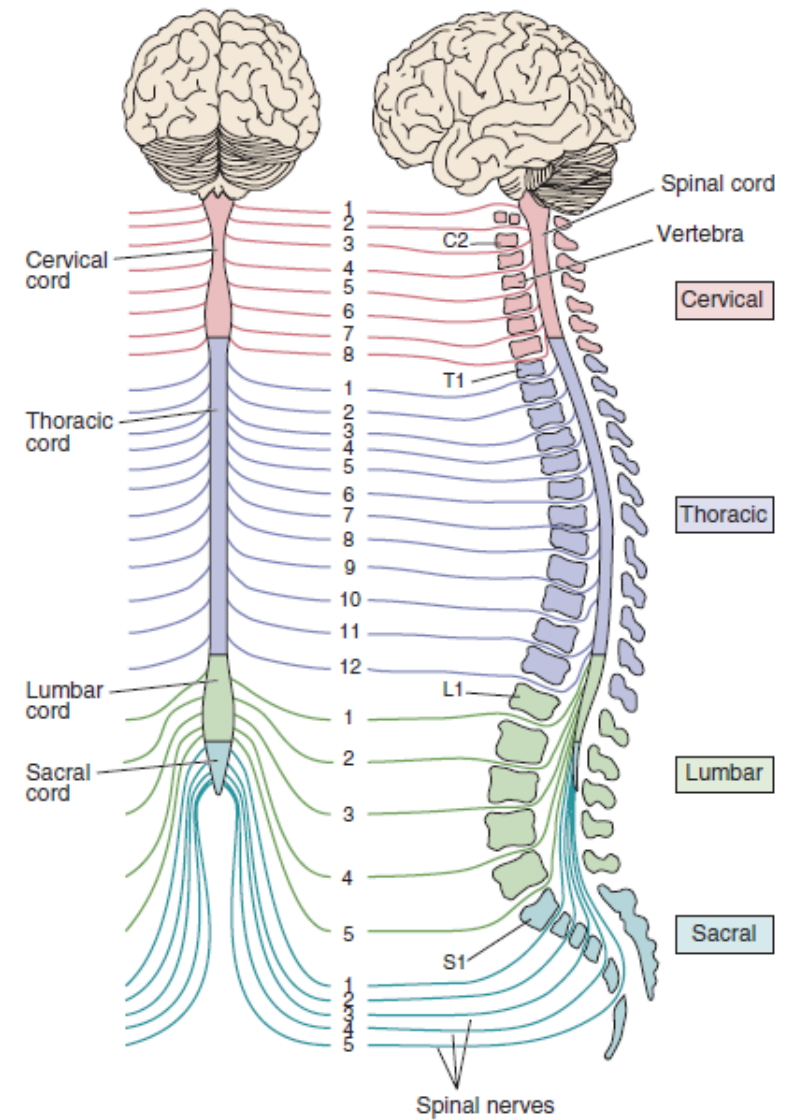
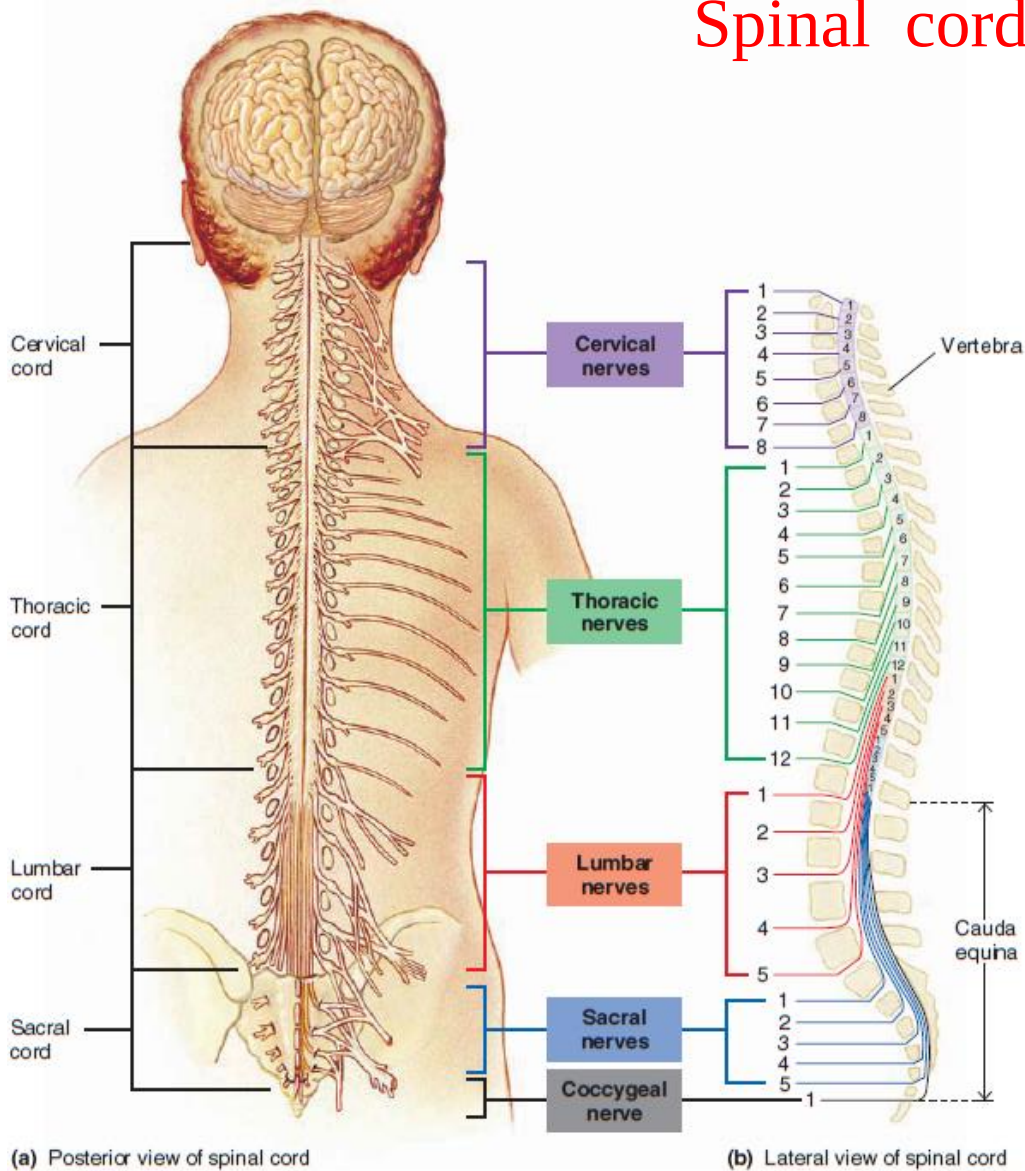
Levels of the Nervous System



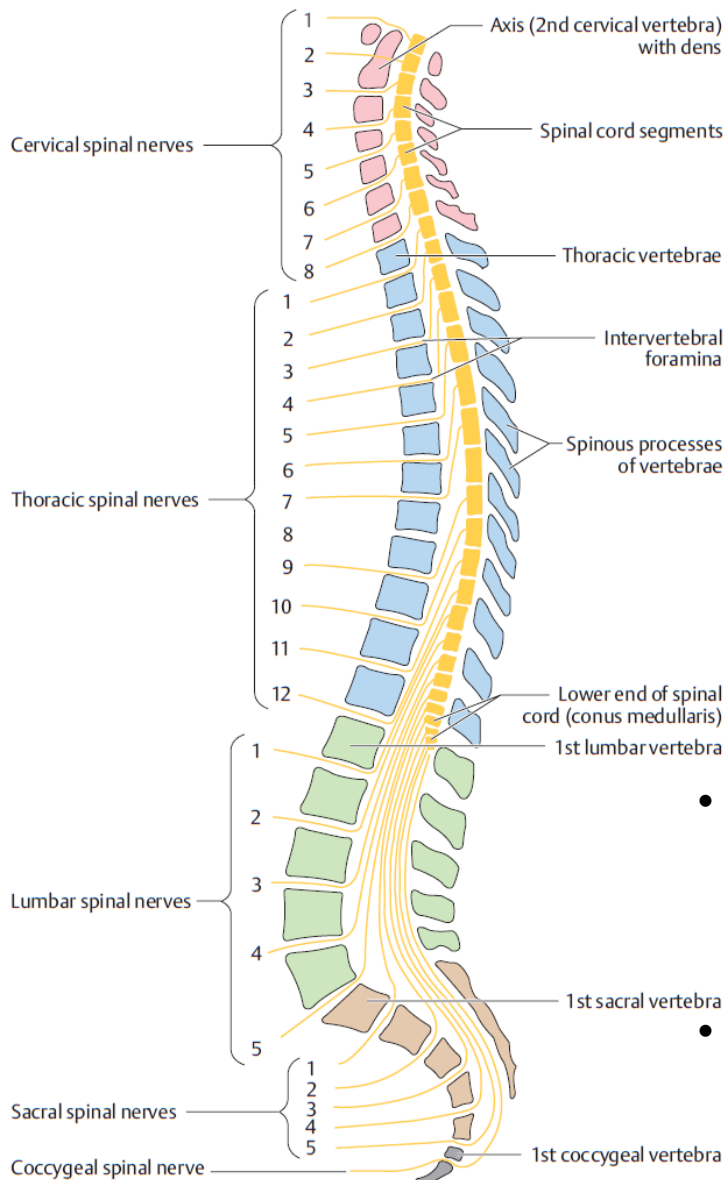




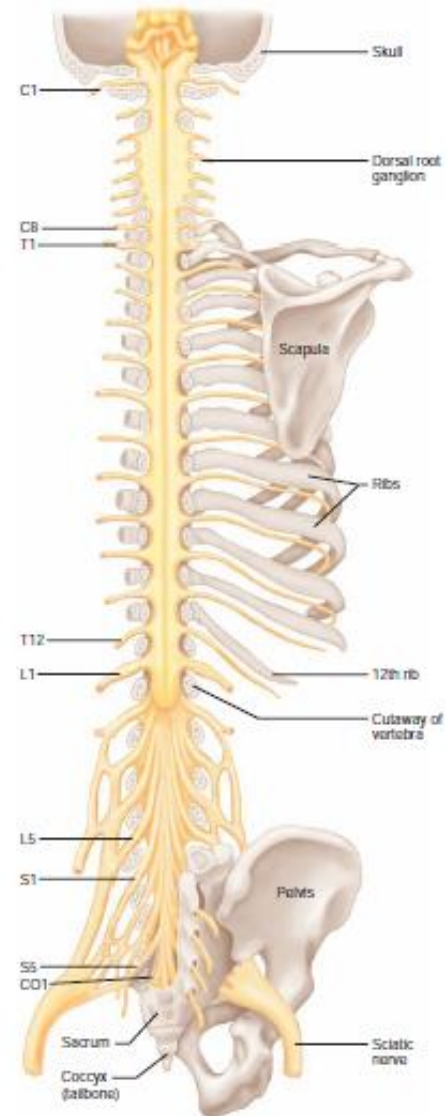
Spinal cord



Spinal cord



- Receives and processes sensory information from the skin, joints, muscles of the limb + trunk.
- Controls movement of the limbs + trunk.

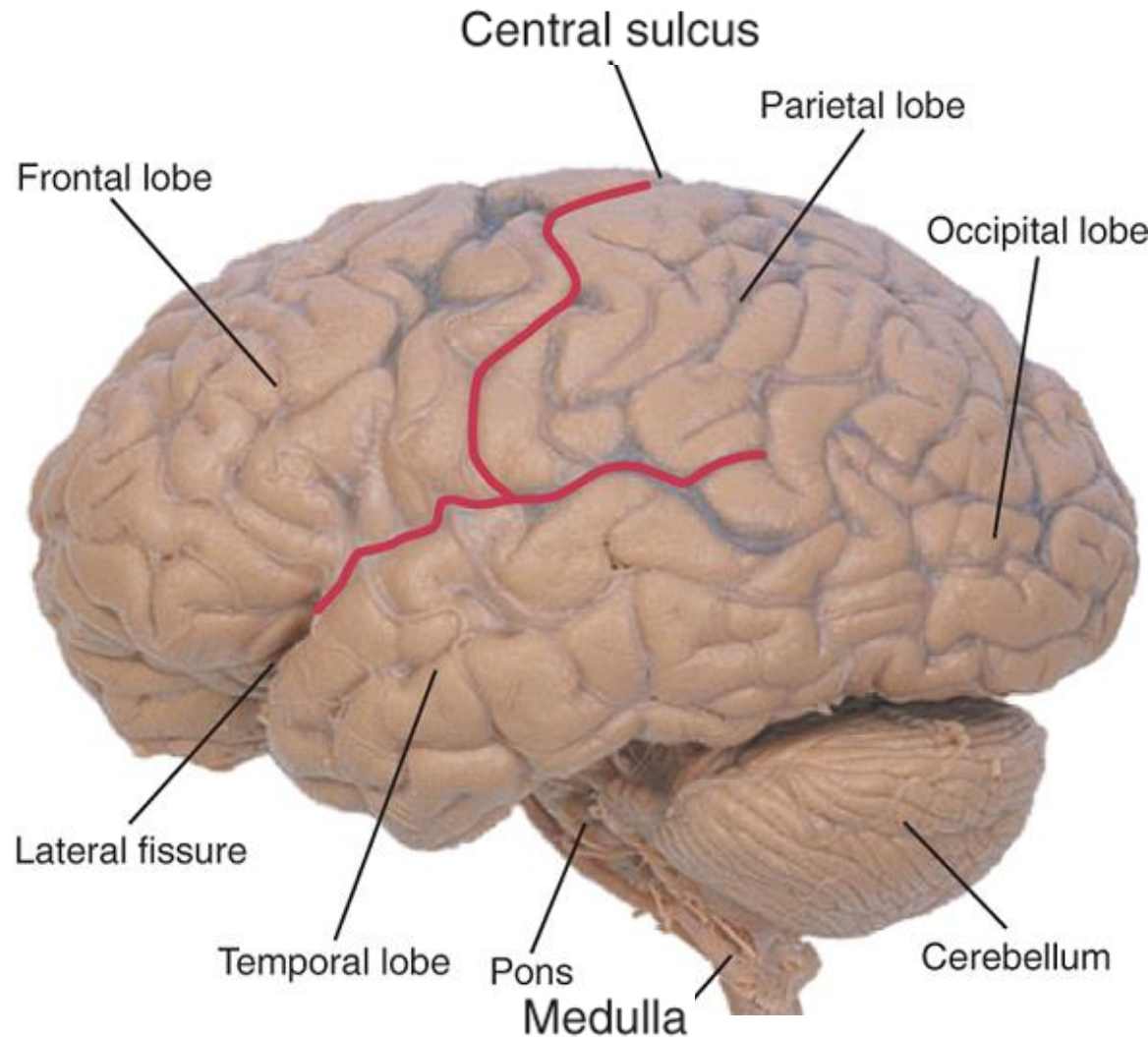


Medulla:

- Vital autonomic functions:
digestion + breathing + HR.

Pons:

- Conveys information about movement from Cerebral hemispheres to the cerebellum.



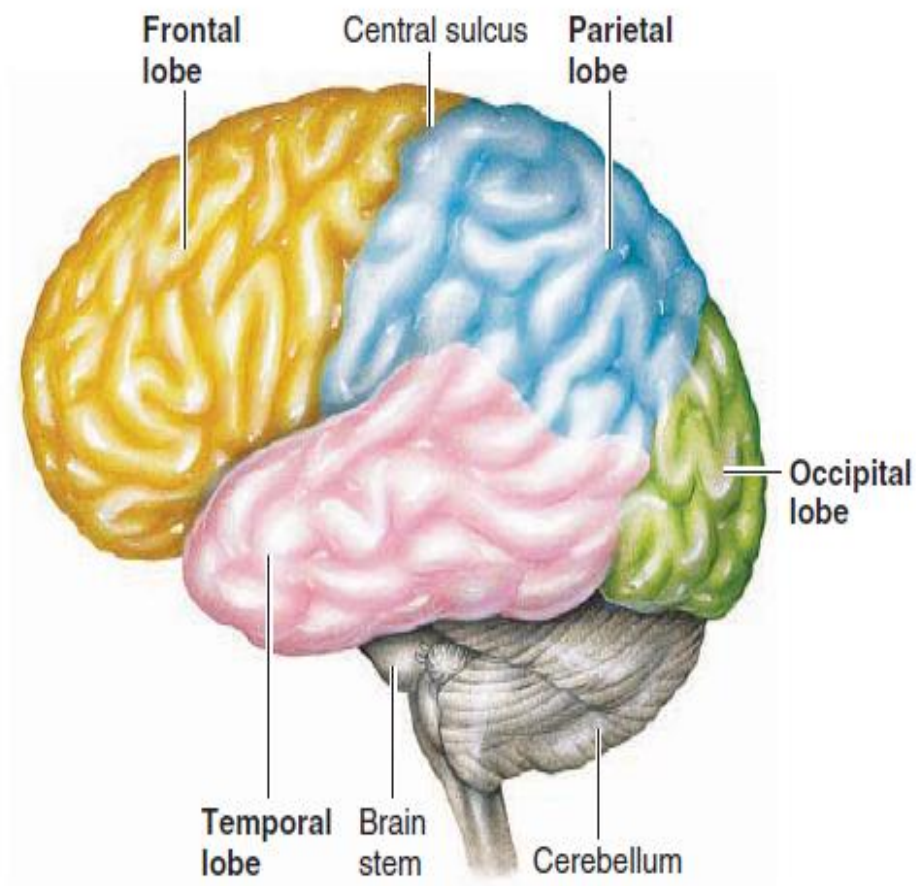
Brain stem:

- Med. oblongata + pons + midbrain.
- Receives sensory information from the skin and muscles of the head.
- Provides motor control for the head musculature.
- Cranial nerve nuclei (*hearing + balance + taste...*)
- Levels of **arousal + awareness** thru RF (SC ↔ Brain).

Midbrain

- Controls sensory + motor functions (eye movements, coordination of visual + auditory reflexes)

Midbrain = Tegmentum + Tectum

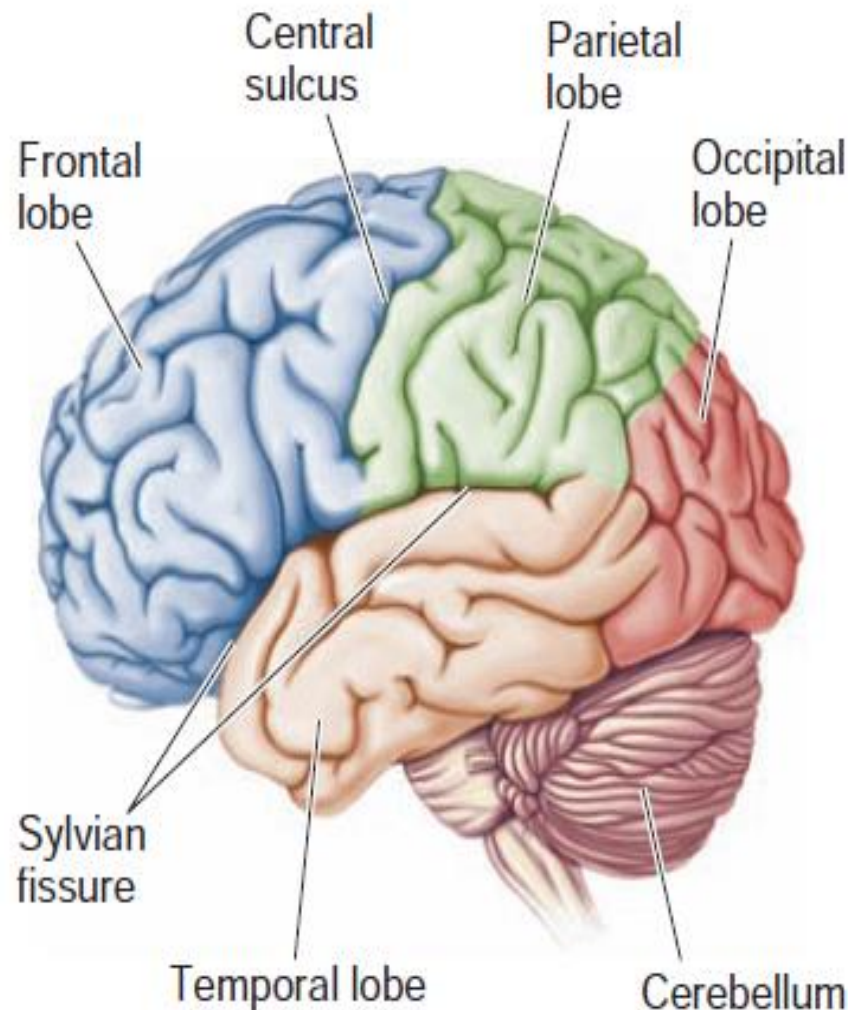


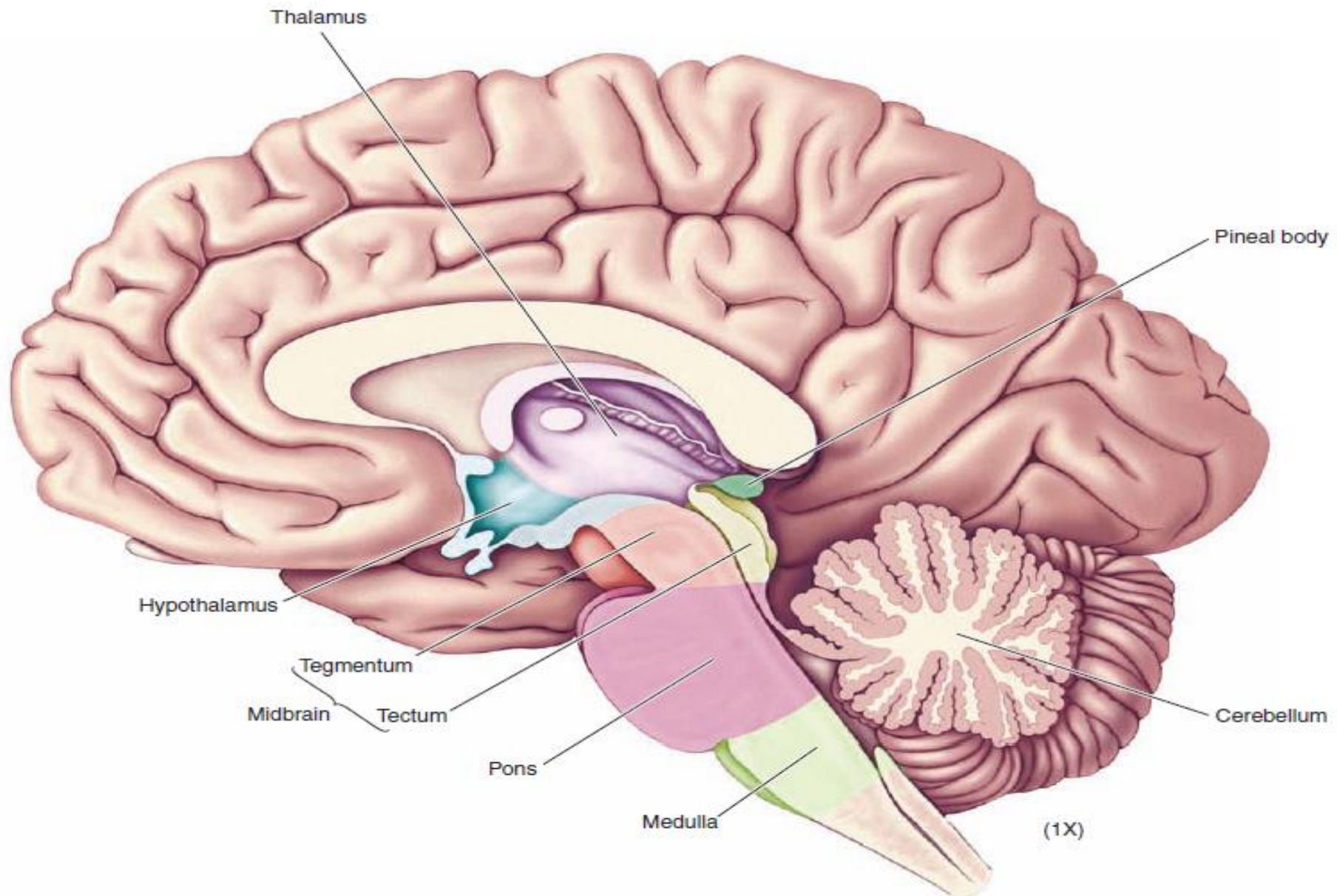
Cerebellum:

- Modulates force + range of movement
- Learning of motor skills.

Diencephalon:

- Relay between the brainstem + cerebral cortex
- **Thalamus** ('secretary' of CC)
- **Hypothalamus** (autonomic + endocrine + visceral functions)

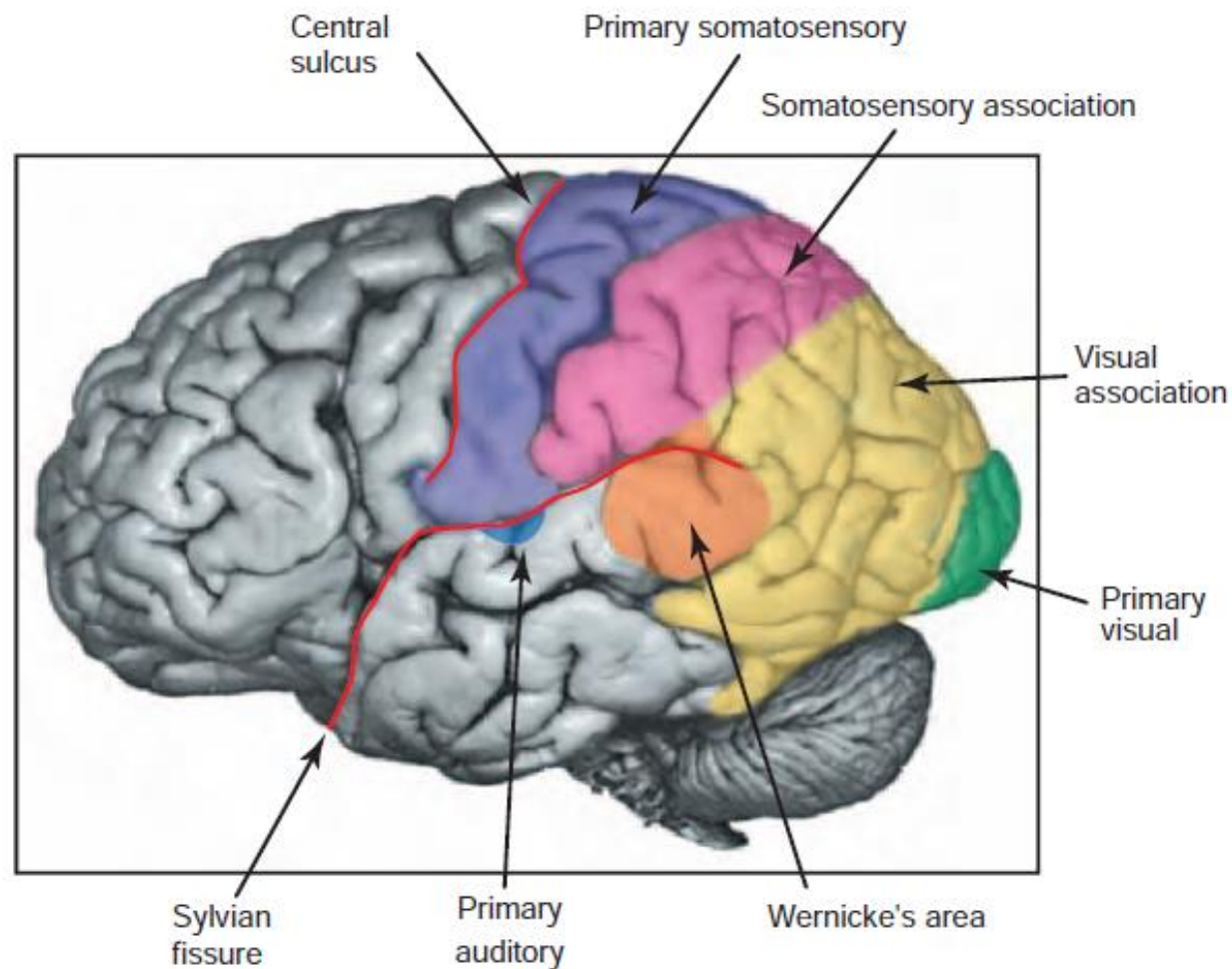




Cerebrum

2 cerebral hemispheres:

- Cerebral cortex
- Basal ganglia
- Hippocampus
- Amygdaloid nuclei



Basal ganglia

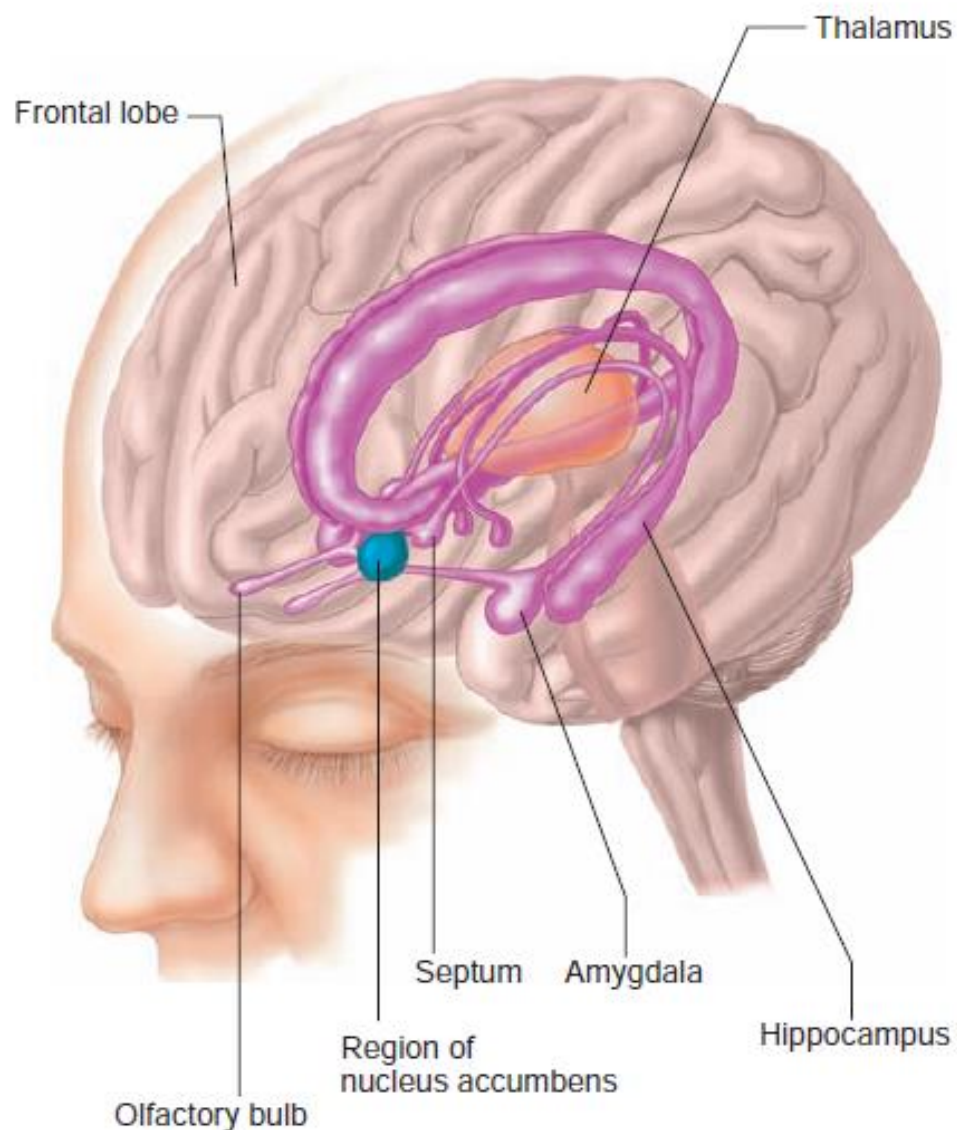
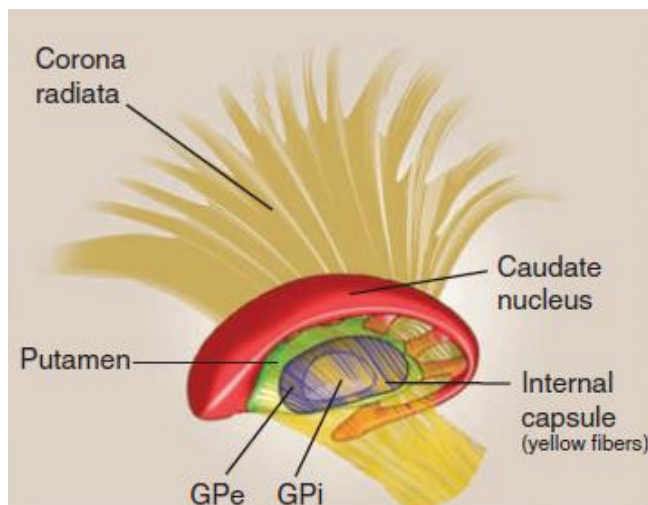
- Motor performance

Hippocampus

- Aspects of memory storage

Amygdaloid nuclei

- Autonomic + endocrine responses of emotional states.



Cerebral cortex

Frontal lobe:

- Short-term memory +
- Planning future actions
- Control of movement

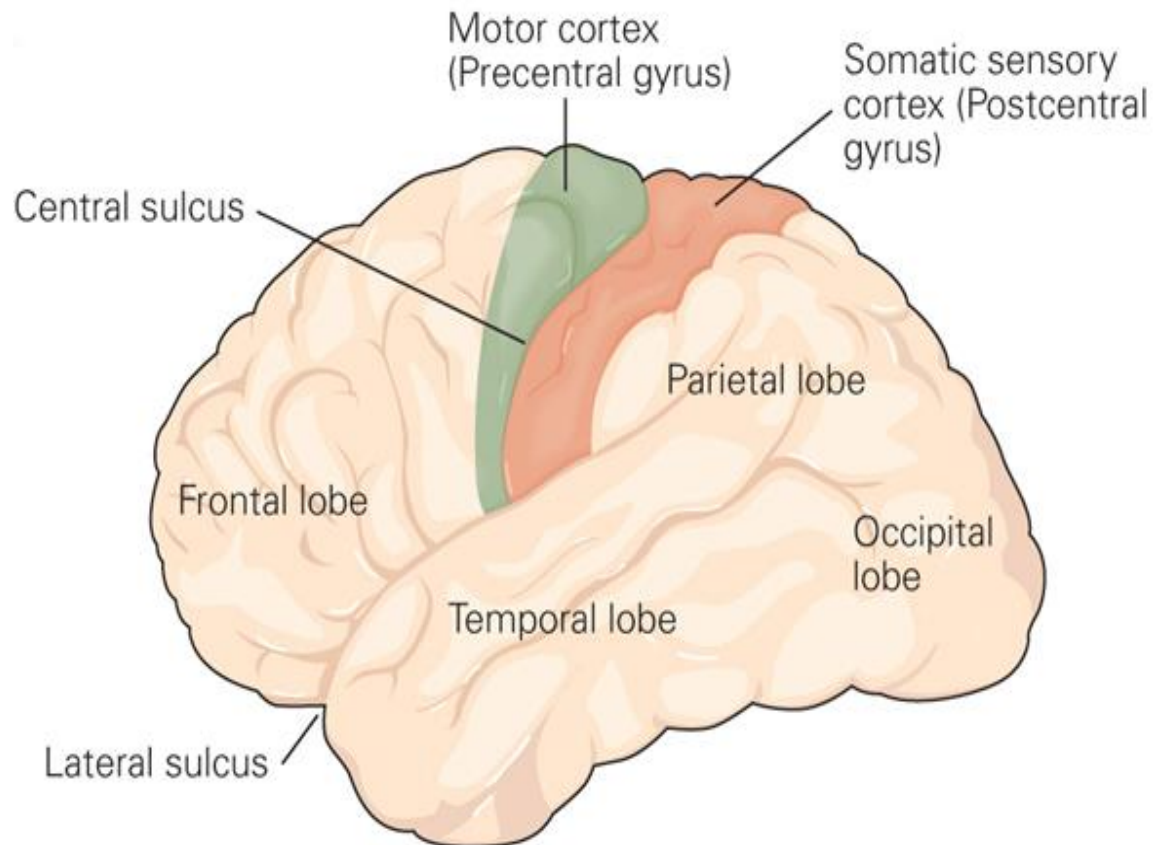
Parietal lobe:

- Somatic sensation
- Forming body image relating to extrapersonal space.

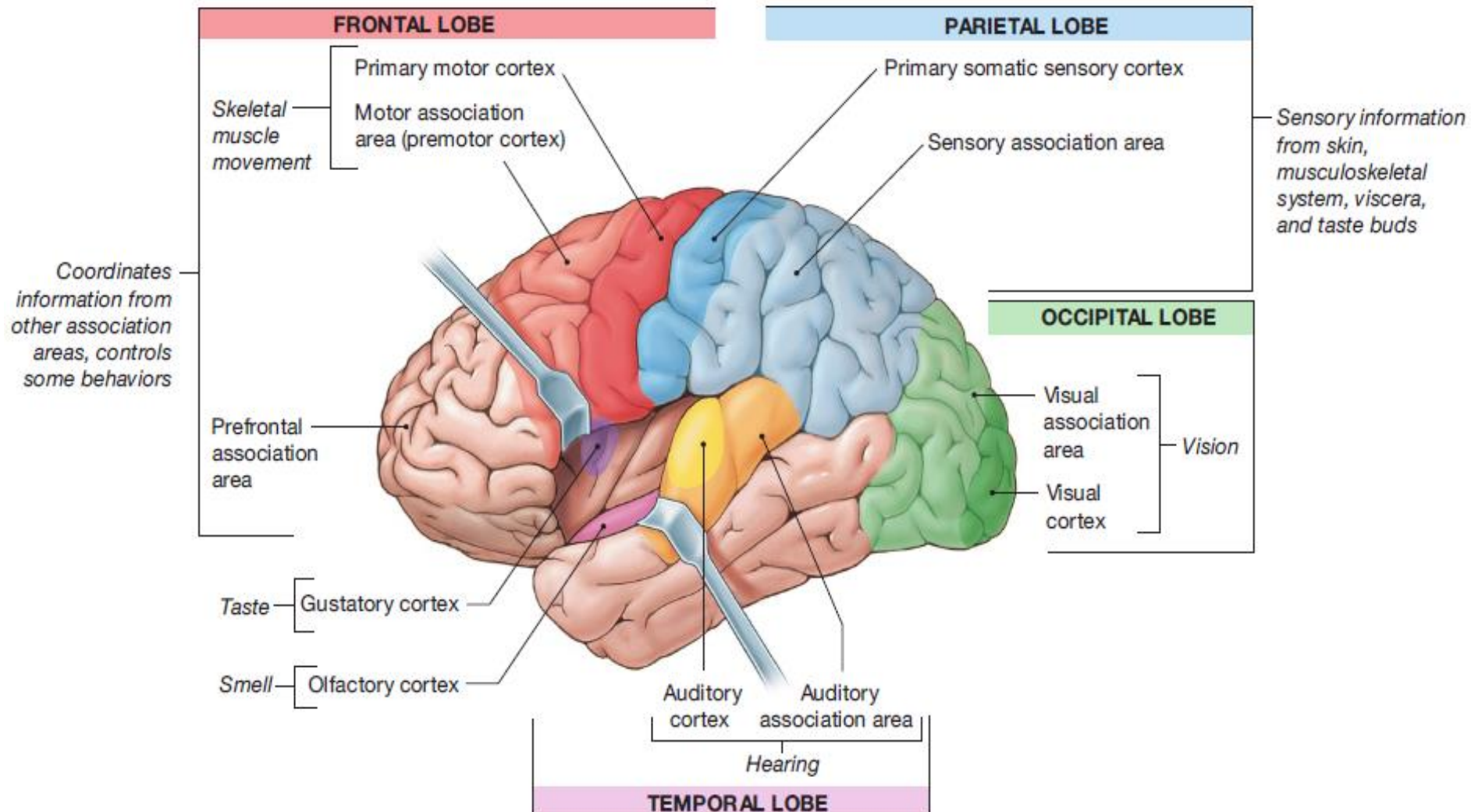
Occipital lobe: vision

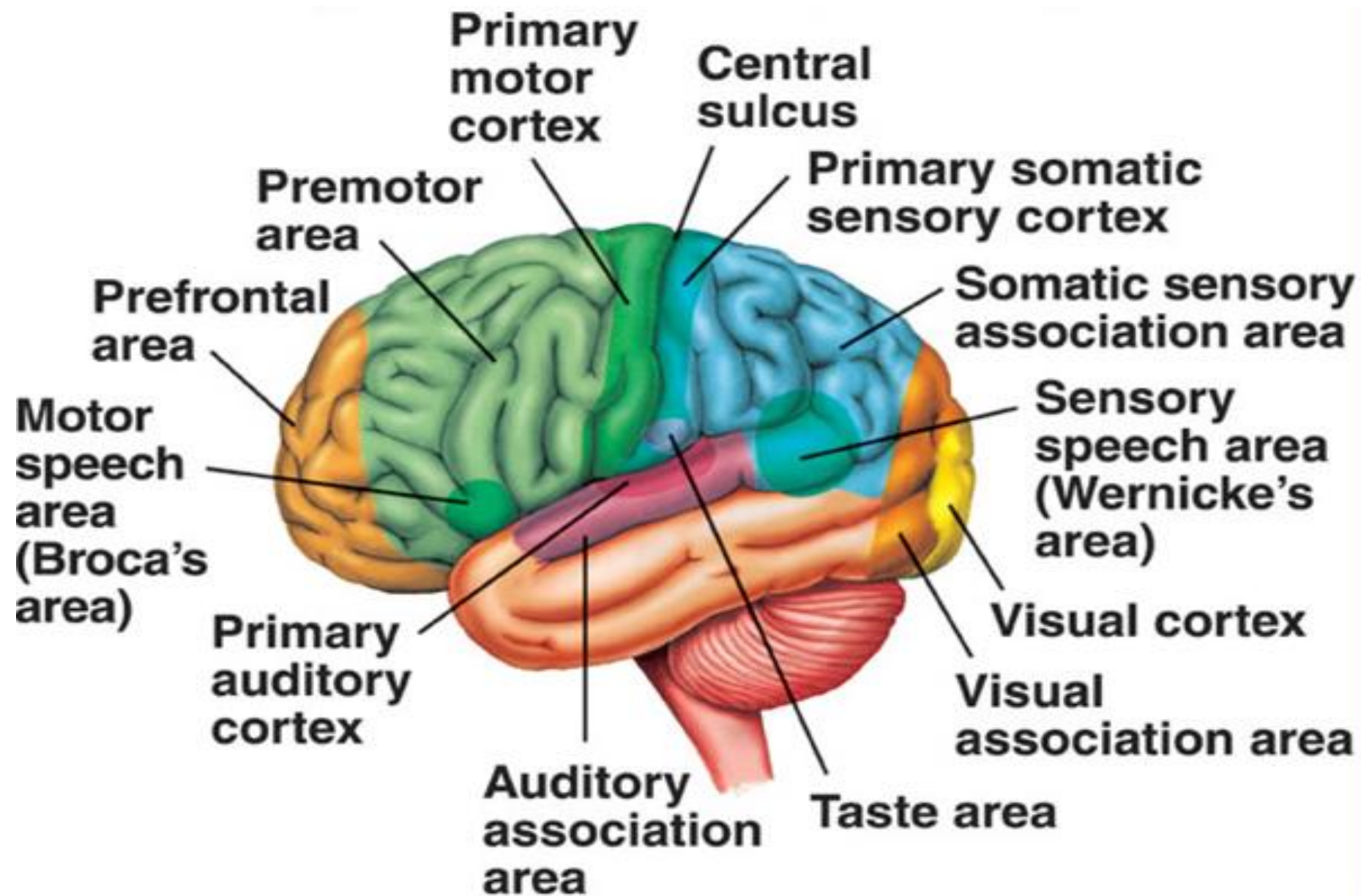
Temporal lobe: hearing

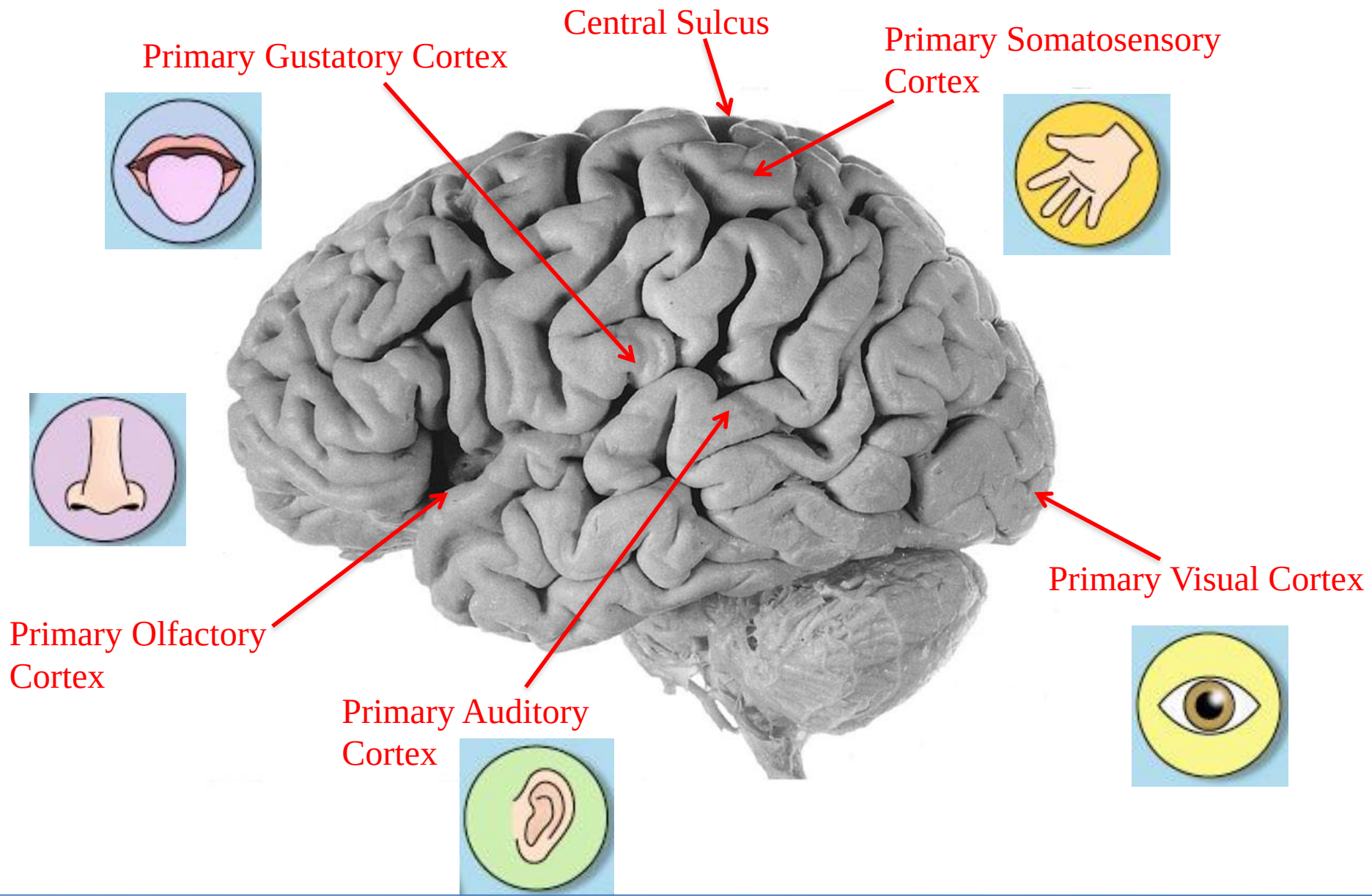
Limbic lobe : **Hippocampus** → Learning + Memory
Amygdaloid nuclei → Emotion



Cerebral cortex







Cerebral cortex

